

DevOps Engineer

Step 4: Learn Kubernetes

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Kubernetes is a declarative platform for managing containerized workloads and services. Its core abstractions include Pods, workload controllers such as Deployments, Services, configuration resources, storage and security. The official documentation describes Pods as the smallest deployable compute object and Services as the abstraction for exposing workloads on a network. ■cite■turn0search1■turn0search22■turn0search23■

The learning objective is to understand the desired-state model: you declare what should run and Kubernetes controllers work toward that state.

Core Concepts — Detailed Breakdown

- Cluster: control plane and worker nodes at a conceptual level.
- Pod: smallest deployable unit; normally one or more tightly coupled containers.
- Deployment: declaratively manages Pods through ReplicaSets and supports controlled rollouts. ■cite■turn0search14■
- Service: stable network endpoint for a changing set of Pods. ■cite■turn0search23■
- ConfigMap/Secret: separate configuration from container images; protect sensitive values appropriately.
- Namespaces: logical isolation and organization.
- Probes: readiness controls traffic eligibility; liveness helps detect unhealthy processes.
- Resources: requests/limits influence scheduling and resource control.
- kubectl: inspect resources, logs, events and rollout state.

What You Should Be Able To Do

- Create a local cluster.
- Deploy a container.
- Expose it with a Service.
- Scale replicas.
- Perform a rolling update.
- Inspect logs and events after an intentional failure.

Hands-on Project / Practice

- Deploy a two-service web application to a local Kubernetes cluster. Write Deployment and Service manifests, add probes, then perform a version update and observe the rollout.

Recommended References

- Kubernetes Concepts — <https://kubernetes.io/docs/concepts/>
- Kubernetes Deployments — <https://kubernetes.io/docs/concepts/workloads/controllers/deployment/>
- Kubernetes Services — <https://kubernetes.io/docs/concepts/services-networking/service/>
- Kubernetes Course — https://www.youtube.com/watch?v=d6WC5n9G_sM

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.